

Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking

Abstract

CCS Concepts

- **Human-centered computing** → **Mixed / augmented reality**;
- **Computing methodologies** → **Rendering**; *Computer vision*.

Keywords

eye tracking, deep learning, computer graphics

ACM Reference Format:

. 2025. Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking. In *Proceedings of SIGGRAPH*. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

2 Related works

To compare our work with existing eye convergence and accommodation tracking, we review related works from two main areas: physical feature-based experimental designs and virtual eye modeling and neural rendering. To clearly show the technical features of previous works, Table 1 categorizes and compares these methods based on their hardware setups, algorithms, operating modes, and tracking accuracy.

2.1 Eye Accommodation Tracking

In stereoscopic and near-eye displays, the **Vergence-Accommodation Conflict (VAC)** [Hoffman et al. 2008] arises when the eyes converge to a virtual 3D depth while the crystalline lens remains focused on the physical display plane. This mismatch can reduce fusion performance and cause visual discomfort. Recent VAC-related depth-perception studies therefore use eye-tracking and controlled optical setups to measure vergence responses under different perceived-depth conditions, highlighting the need for objective and reproducible eye-based measurements in VAC-related experiments [Bontha and Arefin 2024; Sturgeon et al. 2025]. However, vergence alone does not fully describe the accommodative state of the eye. Accommodation is harder to measure directly, especially in compact near-eye systems. Higher-order Purkinje images provide a promising optical cue because they arise from reflections at ocular interfaces, including the crystalline lens surfaces, and vary with the lens geometry during accommodation. Lu et al. [2020] use a multi-camera, multi-LED AR-glasses prototype to track higher-order Purkinje images, extending conventional pupil–corneal-reflection tracking toward accommodation-aware sensing in a near-eye form factor. Ozhan et al. [2023] extend this direction by collecting multi-subject

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH, 979-8-4007-1549-5/2025/08

© 2025 Copyright held by the owner/author(s).
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

Purkinje measurements and combining a Purkinje imaging setup with a **Multi-Layer Perceptron (MLP)**-based regression model to estimate dynamic vergence and accommodation across users.

These works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but they also reveal a practical challenge: P3 and P4 are faint, easily defocused, and highly sensitive to LED–camera–eye geometry. Motivated by this limitation, we first build a controllable 3D optical eye model to analyze the formation, visibility, and stability of P1–P4 under near-eye infrared imaging conditions.

2.2 Reconstructive surgery of the eyeball with adjustable lighting and end-to-end physical simulation

Excerpt for physical capture, for the optical eyes modeling, the main problem including the complicated geometry of outsurface of cornea. For example, eyenerf propose a hybrid representation, combine explicit eyeball model to handle high-frequency high-gloss, in the mean time, Training the Nerf model for implicit representation enables the modeling of the surrounding eye area and the complex appearance within the eye [Li et al. 2022]. In addition,

Khademi et al. [2026] highlight the importance of near-field illumination to eyes modelling this use hybrid mesh + NeRF Rebuild the ocular system with reflective light capabilities and support the common near-field point lights and structured/fringe projectors [Khademi et al. 2026].

Differentiable optical design provides a useful reference for future optimization of the LED–camera layout in our system. s Na et al. [2024] integrates fluidic lens geometry, differentiable ray tracing, and neural reconstruction into a single end-to-end pipeline, allowing the loss to be back-propagated to lens design parameters [Na et al. 2024]. Similarly, Yang et al. [2024] combines ray tracing with wave propagation to jointly optimize refractive-diffractive optics and the reconstruction network [Yang et al. 2024]. These works suggest that optical simulation can be part of the design loop; in our case, future optimization could use the brightness, sharpness, separation from P1, and field-of-view coverage of P3/P4 as objectives for LED and camera placement.

In real captures, higher-order Purkinje images, especially P3 and P4, are relatively faint and sensitive to defocus and sensor noise. Related enhancement and adaptation methods suggest possible post-processing tools: unsupervised anti-noise computational imaging reconstructed 64×64 images at 20 fps with only a 5% sampling ratio [Zhai et al. 2022], diffusion-based style transfer reduced synthetic-to-real FID to 54.56 compared with 65.31 and 73.26 for prior baselines [Chigot et al. 2025], and physics-informed diffusion reduced physical residuals by about two orders of magnitude in a Darcy-flow task and by 33%–35% in topology optimization [Bastek et al. 2025].

Table 1: Comparison of different eye and accommodation tracking methodologies. Accuracy is categorized into High (High), Mid (Mid, representing accuracy between 0.5° and 1.5°), and Low (Low).

No.	Ref.	Camera	Illum.	Method	Algorithm	Target	Mode	Accuracy
1	[Cornsweet and Crane 1973]	Multiple	IR	DPI	Geometric	Vergence	Desktop	Mid
2	[Morimoto et al. 2002]	Single	IR	PCCR	Geometric	Vergence	Desktop	Mid
3	[Sturgeon et al. 2025]	Multiple	IR	PCCR	Geometric	Vergence	Near-eye	Mid
4	[Bontha and Arefin 2024]	Multiple	IR	Mocap	Geometric	Vergence	Near-eye	Low
5	[Lu et al. 2020]	Multiple	IR	Purkinje	Learning	Both	Near-eye	High
6	[Ozhan et al. 2023]	Single	IR	Purkinje	Hybrid	Accom.	Desktop	High
7	[Nair et al. 2020]	Single	None	Ray Tracing	Rendering	Vergence	Synthetic	Mid
8	[Li et al. 2022]	Multiple	Visible	NeRF	Learning	Vergence	Near-eye	Low
9	[Khademi et al. 2026]	Multiple	Visible	NeRF	Learning	Vergence	Near-eye	Low

3 Background

3.1 Optical basis of Purkinje images and VAC

In natural vision, vergence and accommodation are coupled: the eyes rotate toward a target while the crystalline lens changes optical power to focus it on the retina. In AR/VR displays, virtual depth cues may drive vergence to a distance different from the display focal plane, causing the vergence-accommodation conflict (VAC). As holographic and metasurface-waveguide AR displays move toward more realistic 3D imagery, measuring both vergence and accommodation becomes important for evaluating depth perception and visual comfort [Gopakumar et al. 2024].

Conventional eye trackers estimate gaze or vergence from pupil, iris, limbus, or corneal-glint features, which mainly describe eye rotation and provide limited access to accommodation [Nair et al. 2020]. Purkinje images offer a more direct optical cue: P1/P2 are mainly associated with the cornea, while P3/P4 originate from the crystalline lens surfaces and are related to accommodation [Lu et al. 2020][Ozhan et al. 2023]. Because P3/P4 are faint, easily defocused, and sensitive to LED-camera-eye geometry, we first study their formation in a controllable optical eye model before real IR capture.

3.2 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since P1–P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periorcular appearance modeling [???]. Existing Purkinje imaging systems also provide practical guidance for hardware geometry. Both Lu et al. and Ozhan et al. place infrared LEDs close to the camera optical axis to maximize Purkinje visibility while keeping the reflections inside the camera field of view. At the same time, camera positions must balance the visibility of higher-order Purkinje images against eyelid and eyelash occlusions, glint overlap, and

sensor saturation. These observations suggest that LED and camera placement is not arbitrary but jointly determines the brightness, separation, and detectability of P1–P4. Once the capture geometry is determined, synthetic rendering provides an efficient way to evaluate different configurations before physical experiments. RIT-Eyes demonstrates that physically based rendering can generate realistic near-eye images under controllable illumination, while EyeNeRF and subsequent hybrid eye representations further show that explicit corneal geometry is necessary for reproducing specular reflections and refraction, whereas implicit neural representations are better suited for modeling complex periorcular appearance and relighting. These observations motivate the use of explicit optical eye models for Purkinje simulation before subsequent NeRF or 3D Gaussian reconstruction, fall outside the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023].

4 Results and Discussions

References

- Jan-Hendrik Bastek, WaiChing Sun, and Dennis M. Kochmann. 2025. Physics-Informed Diffusion Models. In *The Thirteenth International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=tpYeermigp>
- Jaya Surya Bontha and Mohammed Safayet Arefin. 2024. Effort to Replicate Custom-Made Augmented Reality Haploscope. In *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 37–41.
- Estelle Chigot, Dennis G. Wilson, Meriem Ghrib, and Thomas Oberlin. 2025. Style Transfer with Diffusion Models for Synthetic-to-Real Domain Adaptation. *Computer Vision and Image Understanding* 259 (2025), 104445. doi:10.1016/j.cviu.2025.104445
- T. N. Cornsweet and H. D. Crane. 1973. Accurate Two-Dimensional Eye Tracker Using First and Fourth Purkinje Images. *Journal of the Optical Society of America* 63, 8 (1973), 921–928. doi:10.1364/JOSA.63.000921
- Manu Gopakumar, Gun-Yeal Lee, Suyeon Choi, Brian Chao, Yifan Peng, Jonghyun Kim, and Gordon Wetzstein. 2024. Full-Colour 3D Holographic Augmented-Reality Displays with Metasurface Waveguides. *Nature* 629 (2024), 791–797. doi:10.1038/s41586-024-07386-0
- David M. Hoffman, Ahna R. Girshick, Kurt Akeley, and Martin S. Banks. 2008. Vergence-Accommodation Conflicts Hinder Visual Performance and Cause Visual Fatigue. *Journal of Vision* 8, 3 (2008), 33. doi:10.1167/8.3.33
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (2023). doi:10.1145/3592433
- Wesley Khademi, Jogendra Kundu, Yatong An, Alexander Fix, and David Colmenares. 2026. Reconstructing Realistic and Relightable Eyes. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

- Gengyan Li, Abhimitra Meka, Franziska Müller, Marcel C. Bühler, Otmar Hilliges, and Thabo Beeler. 2022. EyeNeRF: A Hybrid Representation for Photorealistic Synthesis, Animation and Relighting of Human Eyes. *ACM Transactions on Graphics* 41, 4 (2022). doi:10.1145/3528223.3530130
- Conny Lu, Praneeth Chakravarthula, Ye Tao, Steven Chen, and Henry Fuchs. 2020. Improved Vergence and Accommodation via Purkinje Image Tracking with Multiple Cameras for AR Glasses. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 320–331. doi:10.1109/ISMAR50242.2020.00058
- Carlos H. Morimoto, Arnon Amir, and Myron Flickner. 2002. Detecting Eye Position and Gaze from a Single Camera and 2 Light Sources. In *Proceedings of the 16th International Conference on Pattern Recognition*, Vol. 4. IEEE, 314–317. doi:10.1109/ICPR.2002.1047459
- Mulun Na, Hector A. Jimenez Romero, Xinge Yang, Jonathan Klein, Dominik L. Michels, and Wolfgang Heidrich. 2024. End-to-End Optimization of Fluidic Lenses. In *SIGGRAPH Asia 2024 Conference Papers*. ACM. doi:10.1145/3680528.3687584
- Nitinraj Nair, Rakshit Kothari, Aayush K. Chaudhary, Zhizhuo Yang, Gabriel J. Diaz, Jeff B. Pelz, and Reynold J. Bailey. 2020. RIT-Eyes: Rendering of Near-Eye Images for Eye-Tracking Applications. In *Proceedings of the ACM Symposium on Applied Perception*. ACM. doi:10.1145/3385955.3407935
- Faik Ozan Ozhan, Ugur Aygun, Afsun Sahin, and Hakan Urey. 2023. Dynamic Accommodation Measurement Using Purkinje Reflections and Machine Learning. *Scientific Reports* 13, 21625 (2023). doi:10.1038/s41598-023-47572-0
- Matthew Sturgeon, Dakota Kenoyer-Healy, Russell Cohen Hoffing, Steven Thurman, and Mohammed Safayet Arefin. 2025. Examining Eye Vergence During Perceived Depth Changes with Eye Tracking System in Optical See-Through Augmented Reality. In *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 1376–1386.
- Xinge Yang, Matheus Souza, Kunyi Wang, Praneeth Chakravarthula, Qiang Fu, and Wolfgang Heidrich. 2024. End-to-End Hybrid Refractive-Diffractive Lens Design with Differentiable Ray-Wave Model. In *SIGGRAPH Asia 2024 Conference Papers*. ACM. doi:10.1145/3680528.3687640
- Xinliang Zhai, Xiaoyan Wu, Yiwei Sun, Jianhong Shi, and Guihua Zeng. 2022. Anti-Noise Computational Imaging Using Unsupervised Deep Learning. *Optics Express* 30, 23 (2022), 41884–41897. doi:10.1364/OE.470767